

Examiner reports

Q1.

Many candidates produced fully correct and succinct solutions here. Others barely scored any marks, as they seemed unaware of the variable nature of the acceleration and applied the equations of constant acceleration. A few started correctly, but made errors in separating variables prior to integration.

Q2.

Part (a) was frequently done well, with the most common error in part (a)(ii) being the introduction into the equation of motion of an additional force, usually the weight of the motorcycle and rider. Part (b) seldom produced assumptions backed by sound reasons. The majority of responses referred to information given in the question. In part (c), the most convincing solutions were based on the algebraic link between the force and the radius. There was a tendency in part (c) to assume that the magnitude of the frictional force could be altered by changing the radius of the circle, rather than the necessity to adjust the radius to compensate for the force.

Q3.

This was an unusual question and produced many pleasing responses, showing good understanding of the law of static friction. Part (a) was generally done very well. Part (b), as expected, proved more discriminating, with a significant proportion stating 2.8 N as their answer, while some gave negative frictional forces. Part (c) produced many good solutions.

Q4.

Diagrams were mostly very good with only occasional lapses in the direction of the normal reaction force or in poor labelling of forces. Part (a) proved popular and successful for most candidates. In part (b)(i) a number failed to appreciate that the forces involved included the weight component, but many went on to be successful in part (ii).

Q5.

There were some excellent solutions to this question, which was intended to discriminate. The majority knew and attempted to apply the required principles but some were let down by poor algebraic skills in parts (a) and (b). Part (c) showed limited understanding of the equilibrium of the peg. Part (d) was mostly done well, but some failed to appreciate the need to double the distance found when considering the motion of one particle. It should be noted that in a totally algebraic question, candidates should be aware that a numerical value for g is inappropriate. However, the mark scheme treated candidates who used a numerical value sympathetically where possible.

Q6.

This question produced a variety of responses. Parts (a)(i) and (ii) were mostly answered well, although some showed all the forces acting rather than those on particle A as requested. Part (a)(iii) produced irrelevant work, as many failed to appreciate the equilibrium situation, and produced equations of motion as if a dynamic situation existed. Part (a)(iv) proved testing, but many showed a good understanding of the forces involved.

Part (b) was generally done well and accurately.

Q7.

For those who were well prepared in the calculus techniques linking the quantities involved, this question proved highly successful. Part (c)(ii) proved discriminating, as expected, with only the most able candidates linking the components of the velocity vector with the given vector in an appropriate way.

Q9.

This question was done quite well, although some did not find the magnitude of the appropriate quantity in part (a), while others conveniently changed signs in the midst of working to obtain a positive answer.

The simplicity of the request in part (b) was not always realised, and in part (c), some could not find the normal reaction force correctly.

Q10.

Many candidates were fully successful in part (a). In part (b), however, complete equations of motion were rare and components were often inappropriately applied: either incorrect or used with inappropriate forces.

Q11.

Parts (a)(i) and (ii) were often well done, although there was some use of the 'straight string' method, and some lack of understanding of relevant forces acting on the particle on the horizontal surface.

Part (a)(iii) proved discriminating as expected, but part (b) revealed a lack of understanding of the two stages of the motion.

Q12.

A surprising number found the static part of this question very demanding, with relatively few correct responses to parts (a)(i), (iii) and (iv). Part (b) proved more successful.

Q13.

Part (a) proved to be highly discriminating, with few fully correct solutions seen. Candidates showed little understanding of the equilibrium situation of the particles. There was confusion as to which forces acted on each particle, and also between horizontal and vertical forces. The law of friction was known and used by some candidates, but it was rare to see competent handling of inequalities.

Part (b) was often started well, although the description of the surface as 'smooth' was not always appreciated by candidates. Some of the algebraic work was unconvincing, but candidates who were able to write down correct equations for the particles usually completed their solution correctly.

Q14.

This question was the least popular and successful on the paper. Many candidates

showed a lack of understanding of the directions in which forces act, although there were fewer instances of including inappropriate forces than in the past. Part (a)(i) was moderately successful but with many sign errors. Part (a)(ii) was not done well, with few candidates able to write down an equation that contained all the relevant forces. Part (b) proved a further test of understanding, this time testing Newton's first Law, of which a significant number of candidates lacked knowledge.

Q15.

Diagrams in part (a) were often good, although there was a tendency to include the acceleration of the moving object on force diagrams. Part (b) was generally well done but there was some confusion with signs in part (c). Most completed part (d) but sometimes lost a mark through failing to give their answer to three significant figures as demanded in the rubric for the paper.

Q16.

Parts (a) and (b) of this question proved very successful. Some candidates had difficulty in appreciating the nature of the motion after the string broke and chose an incorrect acceleration in part (c). Graphs in part (d) were generally good.

Q17.

Most candidates found this question very straight forward, with very many gaining full marks. A few made minor errors. In part (a) some students simply divided 80 by 20 to give a time of 4 seconds.

In part (b), some candidates introduced the weight of the van. In the final part of the question, some candidates used the correct terms, but introduced the wrong signs.

Q18.

This question was answered fairly well by the majority of candidates. On the whole the force diagrams were quite good, but a number of classic errors were observed. For example including a force in the direction of motion and putting the normal reaction vertical. The rest of part (a) was answered very well, with the candidates appearing to have been helped by the printed answers.

The answers to part (b) were much more variable. There were quite a few very good answers, which included clear concise explanations.

Q19.

Part (a) of this question was generally done very well, although some unusual vector manipulation was observed on a few scripts. Several candidates worked with each component and combined their results into a vector, rather than working with vectors throughout their solutions.

In part (b), a large number of candidates found the velocity correctly, but did not then go on to find the speed of the particle.

There were a fair number of good attempts at part (c). Some candidates made errors with the signs when first setting up their equations. There were a few candidates who did not make any significant attempt at this part of the question.

Q20.

The candidates were more successful with this question than had been expected.

Part (a) was quite well answered. A reasonable number of candidates formed the two equations of motion correctly and then went on to obtain the required value. Some candidates did not resolve the weight of the particle on the slope. A small number of candidates did not form an equation of motion for each particle.

Those candidates who realised that the acceleration was zero did well in part (b). There were quite a few candidates who tried to use the acceleration from part (a) when tackling part (b).

Q21.

Parts (a) to (c) of this question were answered well by many of the candidates, but part (d) caused more problems. There were candidates who simply multiplied the mass and the acceleration, ignoring the resistance force. The other error was to obtain 4600 instead of 4200.

Q22.

Part (a) was generally well answered, although some candidates did make errors when resolving.

Part (b) was found to be much more difficult. The majority of the candidates did not realise that they needed to include a component of the weight and the friction force acting in the same direction. Also a number of candidates did not seem to realise that the acceleration was directed down the slope.

Q23.

This question was done very well. The answer that was given did seem to have helped some candidates. In part (c), a few candidates stated that the tension was zero, even though they had done both parts (a) and (b) correctly.

Q24.

This question was generally done well by the majority of candidates, but some experienced problems. In part (b), there were a few candidates who ignored the instruction to form an equation of motion for each particle. A few others made errors in one equation. Part (c) was usually done correctly if the candidates had two equations of motion in part (b). If they did not have these equations, there were often errors in this part. Part (d) was done well using the acceleration given in the question.

Q25.

Part (a) of this question was done very well and posed little difficulty for the vast majority of candidates. Part (c) was a little more demanding. Some candidates simply found the product ma giving the answer 104 N; others included three terms, but made a sign error in their equation which typically produced a final answer of 69 N.

Q26.

The force diagram asked for in this question was drawn fairly well, but the usual errors of forces in the direction of motion and missing forces were present on some scripts. Parts (b) and (c) were done fairly well and it appeared that many 'candidates' had been helped by the printed answer in part (b). There were some candidates who did not realise that they should consider the tension in the rope when finding the normal reaction. Part (d) was found to be very much more demanding, with relatively few correct responses. Candidate typically omitted one term from their equation or failed to solve correctly for T .

Q27.

This question was generally done very well and many candidates gained full marks. In part (c), a few candidates multiplied the mass of the car by 9.8 instead of 1.2.

Q28.

The candidates found this question a little harder than the previous two questions. There were a number of poor force diagrams, some showing a force in the direction of motion. The errors made by candidates were mainly due to incorrect resolving, but some of these were corrected to produce the answers given in the question. A small number stated that the reaction force was 49 N.

Q29.

There were several good solutions to part (a), but some candidates ignored the instruction to use two equations of motion. Some candidates assumed that A would move up the slope and then ran into difficulties with the negative signs that were involved. Many candidates found part (b) difficult. It was very common for them to try to use the acceleration from part (a) instead of setting the acceleration equal to zero. Interestingly, some candidates who made little or no progress with part (a) did complete part (b) successfully.

Q30.

In this question, there were problems with the force diagram and resolving correctly. Many candidates were able to find the reaction force correctly, but there were more problems in part (d).

Q31.

This question was generally done very well, especially parts (a) and (b). There were, in the main, only minor errors made on these parts. Part (c) did cause difficulties for a small number of candidates. The main problems were that they either omitted the weight or simply calculated it. A few did use the correct terms in an equation that contained a sign error.

Q32.

This question was done quite well on the whole. The major source of errors was resolving incorrectly. The fact that the friction force was given did help some candidates to work towards the correct answer.

Q33.

This question was done well by quite a few candidates. Many of these found the acceleration as part of their solution to part (a). The candidates who had difficulties were those who did not realise that they needed to use a constant acceleration formula or those who quoted an inappropriate formula as their starting point. A few candidates did have difficulty with part (b), but several who had struggled with part (a) used the printed result to enable them to attempt part (b). Those who had been successful with part (b) normally went on to complete part (c).

Q34.

Part (a) was usually done very well. However there were very few good responses to part (b). The most common error was to try to find the magnitude of the resultant of P and the friction, assuming that the friction force had magnitude 58.8 N. For many candidates their understanding of the friction inequality seemed poor. In part (c), the candidates tended to either find the correct value or to simply give the answer 16 N, which was the product of the mass and the acceleration. Many candidates did part (d) very well, but a few omitted the negative sign from the acceleration. Those candidates who had difficulties usually used an incorrect acceleration.

Q35.

In part (a), there were a number of candidates who simply seemed to think that the acceleration was given by distance over time. Many of these candidates also gave the units as m s^{-1} , which further confirmed that their approach was incorrect. However their attempts at part (b) were very much better, probably helped by the printed answer. There were also a number of good solutions to part (c). The most common error in part (c) was to ignore the acceleration when calculating the friction force.